

BASIS FOR DECISION MEMO

Processor: Jeff Fischer
Permit No. MI0020257

Designated Name: **Cadillac WWTP**

Date: May 14, 2008
PID #: 73449.1

Monitoring Point 001A: Authorization to discharge 3.2 MGD of treated municipal wastewater from Monitoring Point 001A through Outfall 001 via a pipeline. Outfall 001 discharges to the Clam River.

<u>Parameter</u>	<u>Maximum Limits for Quality or Concentration</u>				<u>Maximum Limits for Quality or Concentration</u>				<u>Frequency of Analysis</u>	<u>Sample Type</u>	<u>Basis for Limits</u>
	<u>Monthly</u>	<u>7-day</u>	<u>Daily</u>	<u>Units</u>	<u>Monthly</u>	<u>7-Day</u>	<u>Daily</u>	<u>Units</u>			
Flow	(report)	---	(report)	MGD	---	---	---	---	Daily	Report Total Daily Flow	PWJ
Carbonaceous Biochemical Oxygen Demand (CBOD5)											
May 1 – Nov 30	190	270	---	lbs/day	7	---	10	mg/l	5x/Week	24-Hr Composite	WQBEL
Dec 1 – Mar 31	670	910	---	lbs/day	25	---	34	mg/l	5x/Week	24-Hr Composite	AB
<i>Apr 1 – Apr 30</i>	<i>530</i>	<i>800</i>	<i>---</i>	<i>lbs/day</i>	<i>20</i>	<i>---</i>	<i>30</i>	<i>mg/l</i>	<i>5x/Week</i>	<i>24-Hr Composite</i>	<i>see notes</i>
Apr 1 – Apr 30	450	670	---	lbs/day	17	---	25	mg/l	5x/Week	24-Hr Composite	WQBEL
Total Suspended Solids											
May 1 – Nov 30	530	800	---	lbs/day	20	30	---	mg/l	5x/Week	24-Hr Composite	PWJ
Dec 1 – Apr 30	800	1200	---	lbs/day	30	45	---	mg/l	5x/Week	24-Hr Composite	STS
Ammonia Nitrogen (as N)											
May 1 – Sep. 30	24	53	---	lbs/day	0.9	---	2.0	mg/l	5x/Week	24-Hr Composite	WQBEL
<i>Oct 1 – Nov. 30</i>	<i>51</i>	<i>53</i>	<i>---</i>	<i>lbs/day</i>	<i>1.9</i>	<i>---</i>	<i>2.0</i>	<i>mg/l</i>	<i>5x/Week</i>	<i>24-Hr Composite</i>	<i>see notes</i>
Oct. 1 – Nov. 30	24	53	---	lbs/day	0.9	---	2.0	mg/l	5x/Week	24-Hr Composite	WQBEL
Dec 1 – Mar 31	110	---	---	lbs/day	4.2	---	---	mg/l	5x/Week	24-Hr Composite	AB
Apr 1 – Apr 30	150	---	---	lbs/day	5.5	---	---	mg/l	5x/Week	24-Hr Composite	AB
Total Phosphorus (as P)											
May 1 – Mar 31	13	---	---	lbs/day	0.5	---	---	mg/l	5x/Week	24-Hr Composite	WQBEL
Apr 1 – Apr 30	17	---	---	lbs/day	0.65	---	---	mg/l	5x/Week	24-Hr Composite	WQBEL
Fecal Coliform Bacteria	---	---	---	---	200	400	---	cts/100 ml	5x/Week	Grab	WQS
<i>Lindane</i>	---	---	---	---	<i>(report)</i>	---	---	<i>µg/l</i>	<i>Quarterly</i>	<i>24-Hr Composite</i>	<i>see notes</i>
<i>Total Cadmium</i>	---	---	---	---	<i>(report)</i>	---	---	<i>µg/l</i>	<i>Quarterly</i>	<i>24-Hr Composite</i>	<i>see notes</i>
<i>Total Copper</i>	---	---	---	---	<i>(report)</i>	---	---	<i>µg/l</i>	<i>Quarterly</i>	<i>24-Hr Composite</i>	<i>see notes</i>
Acute Toxicity	---	---	---	---	---	---	(report)	TU _A	Monthly	24-Hr Composite	PWJ

Parameter	Maximum Limits for Quality or Concentration				Maximum Limits for Quality or Concentration				Frequency of Analysis	Sample Type	Basis for Limits
	Monthly	7-day	Daily	Units	Monthly	7-Day	Daily	Units			
Chronic Toxicity	---	---	---	---	(report)	---	(report)	TU _c	Monthly	24-Hr Composite	see notes
Chronic Toxicity	---	---	---	---	1.1	---	---	TU _c	Monthly	24-Hr Composite	WQBEL
Total Selenium	---	---	---	---	---	---	(report)	µg/l	Quarterly	24-Hr Composite	WQC
Available Cyanide	---	---	---	---	---	---	(report)	µg/l	Quarterly	Grab	WQC
Total Zinc	---	---	---	---	---	---	(report)	µg/l	Quarterly	24-Hr Composite	WQC
Total Mercury	(report)	---	---	lbs/day	(report)	---	---	ng/l	Quarterly	Grab	PWJ
Total Mercury	0.0008	---	---	lbs/day	30	---	---	ng/l	Monthly	Grab	see notes
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NOTES:

Changes from Last Permit:

The Total Copper, Total Cadmium, and Lindane monitoring requirements were removed from the permit. Analytical data demonstrate these parameters no longer have the potential to exceed water quality standards. New monitoring requirements for Total Selenium and Available Cyanide were added to the permit, there is concern these parameters may have the potential to exceed water quality standards. Also, a final effluent limit for Chronic toxicity was also included in the reissued permit. More restrictive limitations for CBOD5 and Ammonia Nitrogen, for the time period from May – Nov & May – Sept, respectively were included as was a more restrictive limit for Total Mercury based on the revised mercury permitting strategy promulgated in 2005. Also, the monitoring frequency for total mercury is being reduced to quarterly for this permit cycle because the permittee has maintained the concentration of total mercury below 5 ng/l for the entire last permit cycle.

Remediation Wastewater:

The permittee indicated in their application that there is the possibility they may start accepting remediation wastewater from the LUST clean-up. The superintendent said it is unlikely, but possible. To address this possibility, a condition, Part I.A.f, was included that requires the permit to notify the department in accordance with Part II.C.10, and expresses the right of the department to impose additional requirements as deemed necessary.

PERMIT CONDITIONS:

Pollutant Minimization Program for Total Mercury
Michigan Industrial Pretreatment Program
Residuals Management Program
Additional Monitoring Requirements
Storm Water Pollution Prevention Plan

Monthly Operating Reports

Untreated or Partially Treated Sewage Discharge Requirements

Remediation Wastewater

Limit Change Key

Normal Type = existing requirement – carried over from previous version

Bold Type = new requirement – not in previous version

Italic = deleted requirement – not carried over from previous version

Basis for Limits Key

WQBEL - Water Quality Based Effluent Limit
STS – Secondary Treatment Standards
WQC - Water Quality Concerns
WQS - Water Quality Standards
WQV – Water Quality Variance
PWJ - Permit Writer's Judgment
AB - Antibacksliding